7.2 Multiplying Polynomials

Name _____ Period ____ Date _____

Algebra I 7.2 (30-minute plan) · Student sheet

I can…

- multiply a monomial and a polynomial.
- multiply two binomials using a table, the Distributive Property, or FOIL.
- multiply a binomial and a trinomial.

Vocabulary · fill in the missing word

Distributive Property — a(b + c) = ab + ac: _____ every term inside by the term outside

FOIL Method — First, Outer, Inner, Last: the four products when _____ two binomials

Area model (table) — a _____ split into boxes, one for each pair of terms

Part 1 · Notes and examples (fill in each note, then work the example)

Monomial × polynomial: To multiply a monomial and a polynomial, _____ the monomial to every term.

Product of Powers: When I multiply powers with the same base, I _____ the exponents.

Example 1

Find the product.

$-3x^2(2x^3 - 4x + 1)$

Example 2

Use a table to find the product.

$(3x - 2)(x + 5)$

FOIL stands for _____, _____, _____, _____.

Binomial × trinomial: A binomial times a trinomial gives _____ products before I combine _____.

Example 3

Use FOIL to find the product.

$(2y - 3)(4y - 1)$

Part 2 · Partner practice

Partner practice

Find the product.

$(x - 6)(x + 2)$